

**FIGMA LINK:**

<https://www.figma.com/design/xLJPxXaTtV1cywIZBTKfJN/cart?node-id=4-3&t=uvXDWXfnVV124Ptc-1>

## **Unbox – E-Commerce Prototype Mobile Application**



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## ABSTRACT

*Unbox* is an e-commerce mobile application with a carousel feature designed to improve the overall user experience by providing a clean, intuitive, and visually appealing interface.

The main feature of *Unbox* is its carousel-style menu inspired by modern social media applications, allowing users to explore products through a smooth and interactive scrolling interface. This approach reduces visual clutter while enhancing product discovery and engagement. By applying user-centered design principles and optimizing the interface for mobile devices, the application creates a more seamless and efficient shopping experience.

Overall, *Unbox* design can improve usability and provide a better alternative to traditional e-commerce applications. By its innovative carousel interface making products faster, more engaging, and more enjoyable for users.

# INTRODUCTION

## Background of the project

With the rise of mobile technology, online shopping has become a daily activity for many users. However, traditional e-commerce apps often use grid or list layouts that can feel crowded and overwhelming, making browsing less enjoyable.

To address this, *Unbox* is designed as a mobile e-commerce application with a cleaner and more interactive interface. Its main feature is a carousel-style menu that allows users to browse products through smooth scrolling, reducing clutter and improving engagement.

By focusing on simplicity and user-centered design, *Unbox* provides a more efficient, visually appealing, and enjoyable shopping experience.

## Purpose and Objectives

The main purpose of this project is to design an interactive e-commerce mobile application that improves how users browse and discover products.

The objectives of this project are:

- To design an interactive system using a carousel-style interface for product browsing.
- To create a clean, simple, and visually appealing mobile design influenced by modern social media aesthetics.
- To improve product discovery through smooth and engaging navigation.
- To apply user-centered design principles in the development of the application.

- To provide a more efficient shopping experience compared to traditional e-commerce apps.

## **Scope**

### Included Features:

- Carousel-style browsing of products
- Purchase History
- Dashboard Shortcuts

### Excluded Features:

- Payment and checkout system
- Delivery and tracking features
- Seller or admin management system

## **Target Users/Audience**

The *Unbox* application is designed for:

- Users who shop using mobile apps
- Users who like modern app designs
- Users who like visually appealing interfaces

*Unbox* is designed for users who want to browse products easily, discover items quickly, and enjoy a smooth, visually appealing shopping experience.

## **RELATED SYSTEM REVIEW**

### **Existing Solutions and Their Limitations**

Many popular e-commerce applications such as Shopee, Lazada, and Amazon use grid-based or list-based layouts for product browsing. While these layouts allow users to view multiple products at once, they often result in cluttered interfaces, especially on mobile devices. Users may feel overwhelmed due to the large number of items displayed simultaneously.

Additionally, traditional interfaces rely heavily on search and category filtering, which can limit product discovery. Users may only see items they actively search for, reducing opportunities for exploration and engagement.

Another limitation is the lack of interactive and visually engaging navigation. Most systems prioritize functionality over experience, making browsing less enjoyable.

### **Technologies Used in Similar Projects**

Most modern mobile application prototypes are developed using design and prototyping tools such as Figma. This tool is widely used for creating user interface designs, wireframes, and interactive prototypes without requiring full system development.

Figma allows designers to:

- Create low-fidelity and high-fidelity prototypes
- Design interactive elements such as buttons and navigation
- Simulate user flows and transitions
- Collaborate easily in real time

Compared to full development frameworks, prototyping tools focus more on user experience design and interface testing rather than backend functionality.

## **Justification for the Chosen Approach**

*Unbox* adopts a carousel-style interface inspired by social media platforms to address the limitations of traditional designs.

This approach:

- Reduces visual clutter by displaying one or a few items at a time
- Encourages exploration through swipe-based navigation
- Improves user engagement by mimicking familiar social media interactions

The system design is influenced by Instagram, particularly its use of vertical scrolling, card-based content presentation, and minimalistic interface. These features are widely recognized for improving user engagement and retention, making them suitable for enhancing product discovery in e-commerce applications.

# SYSTEM DESIGN AND DEVELOPMENT

## Requirement Analysis

- **Functional Requirements**

*Unbox* is designed to provide users with an interactive and user-friendly product browsing experience. The system should be able to perform the following functions:

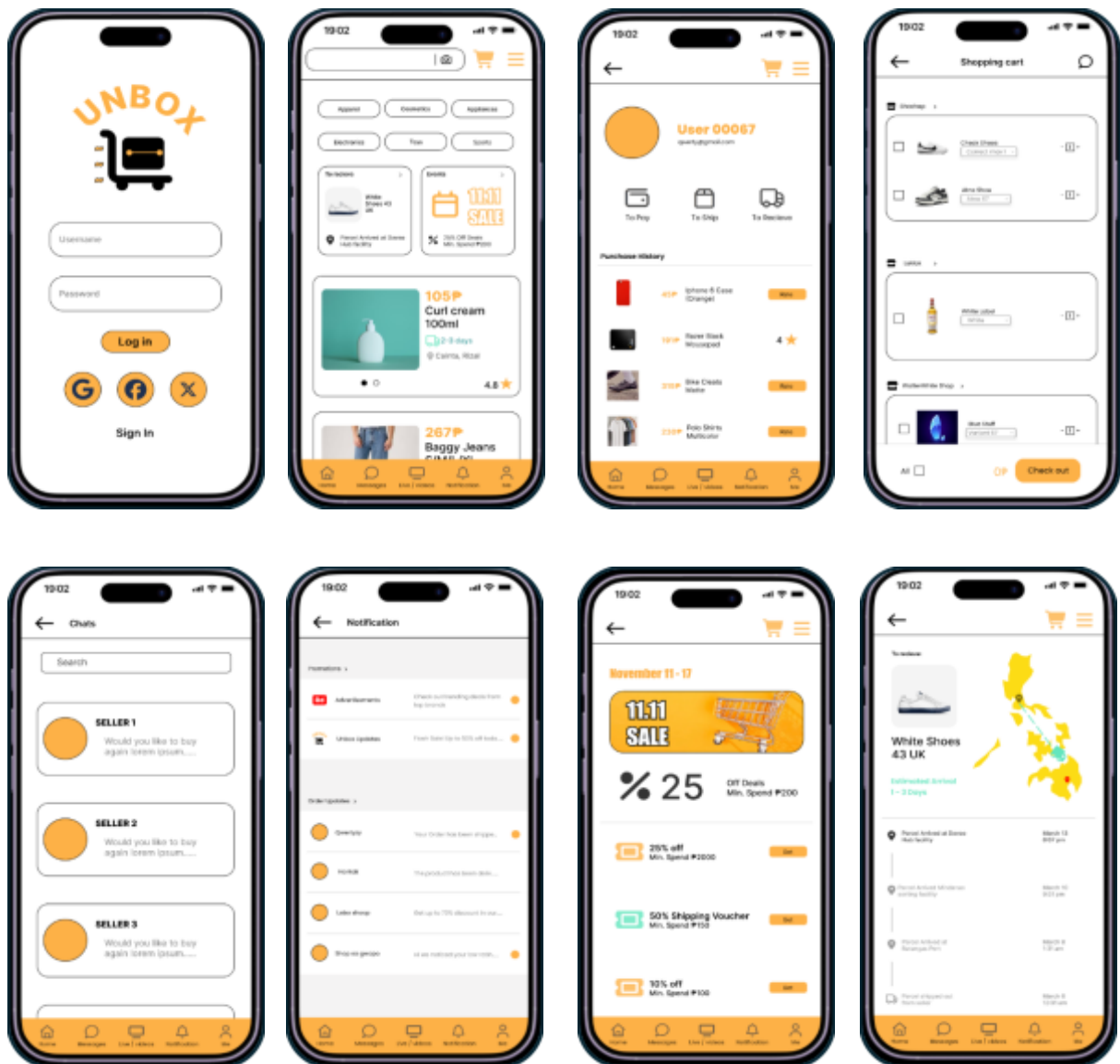
- Allow users to browse products using a carousel-style interface
- Enable smooth scrolling and swipe navigation between products
- Display product details such as name, price, and image
- Provide a dashboard with shortcuts to key sections of the application
- Allow users to view their purchase history
- Enable users to interact with UI elements such as buttons, icons, and menus

- **Non-Functional Requirements**

These requirements focus on the quality and performance of the system:

- Usability: The app should be simple, intuitive, and easy to learn
- Performance: responds quickly with smooth scrolling and transitions
- Accessibility: The design is clear and usable for different types of users
- Consistency: Colors, fonts, and buttons should look uniform on all screens
- Scalability: The design should allow future features like checkout and tracking
- Reliability: The prototype should run smoothly without crashes
- Responsiveness: It should display properly on different mobile screen sizes

## User Interface Design



## Color Scheme

Primary Color: #FDB147 (warm yellow-orange). This warm yellow-orange shade was selected to create a visually appealing and engaging user interface.

Secondary Colors - White, light gray, and black. Neutral colors are used to maintain clarity and balance.



## Typography

*Unbox* uses **Inter** as its primary typeface due to its high readability and suitability for mobile interfaces.

- Font: Inter
- Headings: Bold
- Body: Regular
- Buttons: Medium

- **Implementation Details**

Key functionalities include:

- Carousel interaction using swipe gestures
- Clickable buttons for navigation
- Simulated product pages and dashboard
- Purchase history interface

Screens include:

- Home (carousel view)
- Product details
- Dashboard
- Purchase history

# Testing Methodology

## Unit Testing

Individual components like buttons and navigation were tested to ensure proper interaction.

## Integration Testing

Different screens were tested together to ensure smooth transitions and consistent navigation.

## User Testing

Users were asked to interact with the prototype and provide feedback on usability and design.

## Test Results

| Test Case          | Expected Result   | Actual Result   | Status |
|--------------------|-------------------|-----------------|--------|
| Swipe product      | Smooth transition | Smooth          | Passed |
| Click product      | Opens details     | Works correctly | Passed |
| Navigate dashboard | Opens sections    | Functional      | Passed |

### Issues Identified:

- Minor delays in transitions
- Some buttons needed clearer labels

### Fixes Applied:

- Improved animation timing
- Updated button labels

## **Evaluation**

The system successfully meets its objectives:

- Provides an interactive browsing experience
- Improves usability through clean design
- Enhances product discovery

User feedback showed that the interface is:

- Easy to use
- Visually appealing
- More engaging than traditional layouts

## **CHALLENGES AND LESSONS LEARNED**

### **Challenges**

- Designing a smooth and intuitive carousel interface
- Maintaining consistency across screens
- Balancing simplicity with functionality

### **Solutions**

- Iterative design improvements
- Regular testing and feedback
- Applying UI/UX design principles

### **Lessons Learned**

- Simplicity improves usability
- User feedback is essential
- Good design enhances user experience

## **CONCLUSION AND FUTURE WORK**

### **Conclusion**

Unbox successfully demonstrates how a carousel-based interface can improve the mobile e-commerce experience. The prototype provides a cleaner, more engaging, and user-friendly alternative to traditional layouts.

### **Future Work**

Future improvements may include:

- Adding payment and checkout system
- Integrating delivery tracking
- Implementing user accounts and authentication
- Expanding product categories
- Developing a fully functional backend system